Mathematics: analysis and approaches
Standard level
Paper 1 — markscheme
4.1–4.6 Statistics and probability

13 October 2026

W4ss Academy practice set 2 | **Topic** 4.1–4.6 Statistics and probability | **Ref.** W4SS–MAA–T4–P1–02

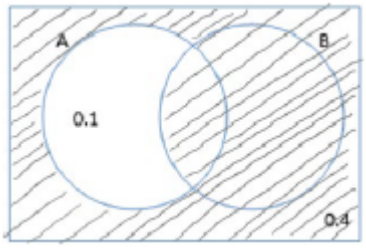
Student name

Paper 1

Instructions to students

- This markscheme is for teachers and for self-assessment after the test.
- **M**: method marks. **A**: answer or accuracy marks, usually after the M mark they depend on. **R**: reasoning marks. **N** (former course): marks for a correct answer with no working.
- Marks in brackets, e.g. **(M1)**, can be implied by correct working that follows. **AG**: answer given in the question, no mark for it. **FT**: follow through from an earlier error.
- Total: **[26 marks]**.

Section A

Question	Marks	Markscheme
1. (a)	**[2]**	valid approach ***(M1)*** *eg* 0.30 − 0.1, $p + 0.1 = 0.3$ $p = 0.2$ ***A1 N2***
1. (b)	**[2]**	valid approach ***(M1)*** *eg* 1 − (0.3 + 0.4), 1 − 0.4 − 0.1 − p $q = 0.3$ ***A1 N2***
1. (c)	**[2]**	valid approach ***(M1)*** *eg* 0.7 + 0.5 − 0.3, $p + q + 0.4$, 1 − 0.1, $P(A' \cup B) = P(A') + P(B) - P(A' \cap B)$  $P(A' \cup B) = 0.9$ ***A1 N2***
2. (a)	**[3]**	attempt to use definition of outlier $1.5 \times 20 + Q_3$ ***(M1)*** $1.5 \times 20 + U \geq 75$ ($\Rightarrow U \geq 45$, accept $U > 45$) OR $1.5 \times 20 + Q_3 = 75$ ***A1*** minimum value of $U = 45$ ***A1***
2. (b)	**[2]**	attempt to use interquartile range ***(M1)*** $U - L = 20$ (may be seen in part (a)) OR $L \geq 25$ (accept $L > 25$) minimum value of $L = 25$ ***A1***
3. (a)	**[1]**	$P(A \cap B) = 0.24$ ***A1***
3. (b)	**[3]**	$P(A \cup B) = 1.1 - P(A \cap B)$ ***(A1)*** $(0 \leq) P(A \cup B) \leq 1$ ***(M1)*** **Note:** This may be conveyed in a clearly labelled diagram or written explanation where $P(A \cup B) = 1$ the minimum value of $P(A \cap B)$ is 0.1 ***A1***

3. (c)	**[2]**	A is a subset of B (so $P(A \cap B) = P(A)$). ***R1*** **Note:** This may be conveyed in a clearly labelled diagram where A is completely inside B, or in a written explanation indicating that $P(A \cap B) = P(A)$ so the maximum value of $P(A \cap B)$ is 0.3 ***A1*** **Note:** Do not award ***R0A1***.
4. (a)	**[2]**	$LQ = \frac{x_1 + x_2}{2}, UQ = \frac{x_4 + x_5}{2}, IQR = \frac{x_4 + x_5 - x_1 - x_2}{2}$ ***M1A1***
4. (b)	**[5]**	$UQ + 1.5IQR = 1.25x_4 + 1.25x_5 - 0.75x_1 - 0.75x_2 \geqslant x_5$ ***M1A1*** Since $1.25x_4 + 0.25x_5 \geqslant 0.75x_1 + 0.75x_2$ due to the ascending order. ***R1*** Similarly $LQ - 1.5IQR = 1.25x_1 + 1.25x_2 - 0.75x_4 - 0.75x_5 \leqslant x_1$ ***M1A1*** Since $0.25x_1 + 1.25x_2 \leqslant 0.75x_3 + 0.75x_4$ due to the ascending order. So there are no outliers for a data set of 5 numbers. ***AG***
4. (c)	**[2]**	For example 1, 2, 3, 4, 5, 6, 100 where $IQR = 4$ ***A1A1***

Please **do not** write on this page.

Answers written on this page
will not be marked.